Vaibhav Patil

Software Developer — C++ • OOP • Multithreading • Windows Desktop • CAD/CAE Visualization
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Professional Summary

Software developer with **1.5 years** at **Tata Consultancy Services** building engineering-oriented applications for **50+ engineers** and **200+ users**. Strong foundation in **C++, object-oriented design, data structures and concurrency concepts**, with hands-on experience in **VTK-based data visualization**, performance optimization and automation. Looking to grow in **C++ desktop and CAD/CAE tooling** (MFC, COM, NX, STAR-CCM+ workflows).

Technical Skills

- **Languages:** C++ (STL, templates, RAII, smart pointers), C, Python, SQL
- **OOP & Design:** Encapsulation, inheritance, polymorphism, SOLID, design patterns (Factory, Observer, Strategy), data structures & algorithms
- **Concurrency:** Threads, mutexes, condition variables, thread pools, race conditions & deadlock avoidance (C++11/17 `<thread>`, `<mutex>`, `<atomic>`)
- **Visualization & CAD/CAE:** VTK; familiar with NX and STAR-CCM+ data workflows (CAD geometry, mesh and simulation result formats)
- **Windows Desktop:** Familiar with MFC (document/view, message maps) and COM fundamentals (interfaces, reference counting)
- **Tools:** Git, Visual Studio, CMake, Linux, CI/CD, PostgreSQL, MySQL
- **Also:** Python (Django, FastAPI, Pandas), React.js, AWS (EC2, Lambda, S3)

Professional Experience

Python Developer, Tata Consultancy Services **Aug 2024 - Present**
Bangalore, India *1.5 Years Experience*

- Designed and delivered **8 modular, object-oriented applications** (Django/FastAPI), improving efficiency by **30%** for **200+ users**
- Built **engineering analysis dashboards** on Python backends with VTK-based visualization support, reducing analysis time by **40%** for **50+ engineers**
- Optimized **PostgreSQL queries** and added Redis caching, cutting API response time by **65%**
- Deployed on **AWS EC2/Lambda** with S3 storage, processing **5GB+ datasets** at **99.9%** uptime
- Automated **12 CI/CD workflows**, saving **160 hours/week**

Education

B.E. in Information Science Engineering, Basaveshwar Engineering College **Dec 2020 - Jun 2024**
GPA: 8.16 *Karnataka, India*

Certifications & Additional Information

- Completed **5 certifications**: Python (Google), Cybersecurity (Cisco), Data Structures
- **Languages:** English (Fluent), Hindi (Native), Kannada (Native)
- **Links:** GitHub: github.com/vaibhav88614 — Portfolio: vaibhav88614.github.io